Results First Draft

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Summary Statistics

Table 1—Summary Statistics, Rainfall Class Samples

	Panel A: WAZ, age ≤10 in 2005 and 2009					
	Mean	Std. dev.	Min.	Median	Max.	Obs.
Outcomes (MxFLS-2, 2005)						
Weight-for-age z-score (WAZ)	-0.30	1.42	-5.07	-0.36	4.51	1,045
Outcome change, 2005 to 2009						
Change in WAZ	0.55	1.41	-4.88	0.46	9.10	1,045
Rainfall classes (CHIRPS)						
2009 rainfall, z	-0.50	0.83	-2.24	-0.39	0.96	1,045
Near normal, $-1 < z < 1$	0.65	0.48	0.00	1.00	1.00	1,045
Moderately dry, $-1.5 < z \leq -1$	0.21	0.41	0.00	0.00	1.00	1,045
Severely dry or worse, $z \leq -1.5$	0.14	0.35	0.00	0.00	1.00	1,045

	Panel B: HAZ and BMIZ, age ≤15 in 2005 and 2009					
	Mean	Std. dev.	Min.	Median	Max.	Obs.
Outcomes (MxFLS-2, 2005)						
Height-for-age z-score (HAZ)	-0.58	1.49	-5.46	-0.63	5.65	2,351
BMI-for-age z-score (BMIZ)	0.36	1.43	-4.95	0.30	4.86	2,310
Outcome change, 2005 to 2009						
Change in HAZ	0.07	1.37	-5.93	0.07	6.35	2,351
Change in BMIZ	0.23	1.31	-5.90	0.17	7.59	2,295
Rainfall classes (CHIRPS)						
2009 rainfall, z	-0.55	0.84	-2.24	-0.53	0.96	2,351
Near normal, $-1 < z < 1$	0.64	0.48	0.00	1.00	1.00	2,351
Moderately dry, $-1.5 < z \leq -1$	0.21	0.41	0.00	0.00	1.00	2,351
Severely dry or worse, $z \leq -1.5$	0.16	0.36	0.00	0.00	1.00	2,351

Notes: This table is for the rainfall class specification, the main model. Obs. counts children. The sample is children with the outcome measured in MxFLS-2 (2005) and MxFLS-3 (2009), under the panel's age cutoff in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Children in wet municipalities (z of 1 or above) are excluded. Panel A uses the WAZ sample, Panel B the HAZ and BMIZ sample. The BMIZ sample is marginally smaller than the HAZ sample shown in Panel B, at 2,300 children. Statistics are at the 2005 baseline, except the change in WAZ, HAZ and BMIZ from 2005 to 2009, the outcome of the rainfall class regressions, and 2009 rainfall. z is 2009 May–October CHIRPS rainfall, the maize growing season (Jessoe et al., 2018), in the child's 2005 municipality in standard deviations from its 1981–2004 mean, grouped into standardized rainfall classes with the SPI cutoffs (McKee et al., 1993). The rainfall class regressions have no controls, as the contdid package does not yet support covariates (Callaway et al., 2024).

Table 2—Summary Statistics, Binary DID Model 3 Samples

	Panel A: WAZ, age ≤10 in 2005 and 2009					
	Mean	Std. dev.	Min.	Median	Max.	Obs.
Outcomes (MxFLS-2, 2005)						
Weight-for-age z-score (WAZ)	-0.31	1.40	-4.92	-0.34	4.51	700
Binary DID treatment (CHIRPS)						
2009 drought (indicator)	0.36	0.48	0.00	0.00	1.00	700
2005 drought (indicator)	0.68	0.47	0.00	1.00	1.00	700
2006 drought (indicator)	0.03	0.17	0.00	0.00	1.00	700
2007 drought (indicator)	0.15	0.36	0.00	0.00	1.00	700
2008 drought (indicator)	0.13	0.34	0.00	0.00	1.00	700
Remittances (MxFLS-1, 2002)						
Household received remittances, proxy (indicator)	0.41	0.49	0.00	0.00	1.00	634
Household (MxFLS-1, 2002)						
Dwelling has electricity (indicator)	0.97	0.16	0.00	1.00	1.00	700
Good floor quality (indicator)	0.85	0.36	0.00	1.00	1.00	700
Good roof quality (indicator)	0.81	0.39	0.00	1.00	1.00	700
Enough ventilation (indicator)	0.57	0.50	0.00	1.00	1.00	700
Sleeping rooms per household member	0.41	0.25	0.11	0.33	2.33	700
Ln (monthly expenditure per capita, pesos), proxy	6.56	0.86	4.21	6.57	11.43	700
Rural locality (indicator)	0.43	0.50	0.00	0.00	1.00	700
Child and mother (MxFLS-2, 2005)						
Age (years, not in regression)	2.52	1.68	0.00	3.00	6.00	700
Male (indicator)	0.53	0.50	0.00	1.00	1.00	700
Fully vaccinated (indicator)	0.39	0.49	0.00	0.00	1.00	700
Mother's height (centimeters)	154.79	6.91	136.00	154.50	181.00	700
Mother's years of schooling	7.40	3.42	0.00	8.00	18.00	700
Mother's age at birth (years)	26.93	6.52	14.00	27.00	59.00	700
Mother employed (indicator)	0.24	0.42	0.00	0.00	1.00	700
Father and birth (MxFLS-1 and MxFLS-2)						
Father's height (centimeters)	166.81	6.98	144.00	167.00	189.40	465
Father's years of schooling	7.51	3.77	0.00	8.00	18.00	569
Father's age at birth (years)	30.74	7.82	16.00	30.00	75.00	581
Father employed (indicator)	0.96	0.19	0.00	1.00	1.00	569
Birth order	2.31	1.68	1.00	2.00	13.00	401
Oldest child (indicator)	0.40	0.49	0.00	0.00	1.00	401
Birth weight (kilograms)	3.26	0.59	1.55	3.20	5.50	523
Delivered at home (indicator)	0.08	0.27	0.00	0.00	1.00	560
Weeks predominantly breastfed	31.98	29.20	0.00	26.09	157.29	542
Iron during pregnancy (indicator)	0.76	0.43	0.00	1.00	1.00	549
Premature, mother reported (indicator)	0.19	0.39	0.00	0.00	1.00	546

	Panel B: HAZ and BMIZ, age ≤15 in 2005 and 2009					
	Mean	Std. dev.	Min.	Median	Max.	Obs.
Outcomes (MxFLS-2, 2005)						
Height-for-age z-score (HAZ)	-0.60	1.47	-5.17	-0.62	5.20	1,755
BMI-for-age z-score (BMIZ)	0.40	1.41	-4.95	0.35	4.86	1,721
Binary DID treatment (CHIRPS)						
2009 drought (indicator)	0.39	0.49	0.00	0.00	1.00	1,755
2005 drought (indicator)	0.66	0.47	0.00	1.00	1.00	1,755
2006 drought (indicator)	0.03	0.17	0.00	0.00	1.00	1,755
2007 drought (indicator)	0.16	0.36	0.00	0.00	1.00	1,755
2008 drought (indicator)	0.14	0.34	0.00	0.00	1.00	1,755
Remittances (MxFLS-1, 2002)						
Household received remittances, proxy (indicator)	0.36	0.48	0.00	0.00	1.00	1,624
Household (MxFLS-1, 2002)						
Dwelling has electricity (indicator)	0.98	0.15	0.00	1.00	1.00	1,755
Good floor quality (indicator)	0.83	0.37	0.00	1.00	1.00	1,755
Good roof quality (indicator)	0.81	0.39	0.00	1.00	1.00	1,755
Enough ventilation (indicator)	0.56	0.50	0.00	1.00	1.00	1,755
Sleeping rooms per household member	0.38	0.23	0.11	0.33	2.33	1,755
Ln (monthly expenditure per capita, pesos), proxy	6.52	0.86	3.75	6.51	12.24	1,755
Rural locality (indicator)	0.46	0.50	0.00	0.00	1.00	1,755
Child and mother (MxFLS-2, 2005)						
Age (years, not in regression)	6.08	3.22	0.00	6.00	12.00	1,755
Male (indicator)	0.50	0.50	0.00	0.00	1.00	1,755
Fully vaccinated (indicator)	0.50	0.50	0.00	0.00	1.00	1,755
Mother's height (centimeters)	154.39	7.14	133.40	154.20	190.00	1,755
Mother's years of schooling	7.13	3.66	0.00	6.00	18.00	1,755
Mother's age at birth (years)	26.80	6.46	14.00	26.00	59.00	1,755
Mother employed (indicator)	0.28	0.45	0.00	0.00	1.00	1,755
Father and birth (MxFLS-1 and MxFLS-2)						
Father's height (centimeters)	166.42	7.15	144.00	166.30	192.00	1,212
Father's years of schooling	7.33	3.92	0.00	7.00	18.00	1,443
Father's age at birth (years)	30.36	7.85	15.00	29.00	75.00	1,482
Father employed (indicator)	0.96	0.20	0.00	1.00	1.00	1,447
Birth order	2.66	1.89	1.00	2.00	13.00	1,392
Oldest child (indicator)	0.32	0.47	0.00	0.00	1.00	1,392
Birth weight (kilograms)	3.27	0.60	0.85	3.25	6.00	1,256
Delivered at home (indicator)	0.12	0.32	0.00	0.00	1.00	1,544
Weeks predominantly breastfed	33.57	33.98	0.00	26.09	260.89	1,357
Iron during pregnancy (indicator)	0.72	0.45	0.00	1.00	1.00	1,366
Premature, mother reported (indicator)	0.14	0.35	0.00	0.00	1.00	1,362

Notes: This table is for the sample of Model 3, the main binary DID with controls. Obs. counts children. The children have the outcome measured in MxFLS-2 (2005) and MxFLS-3 (2009), under the panel's age cutoff in both rounds, and a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Children also have all Model 3 controls observed and are not alone in their interview-month cell. Panel A uses the WAZ sample, Panel B the HAZ and BMIZ sample. The BMIZ sample is marginally smaller than the HAZ sample shown in Panel B, at 1,714 children. Statistics are at the 2005 baseline, except remittances and household variables (2002). Remittances enter Model 5, and the father and birth controls enter Model 4. Both are observed for fewer children. Drought is May–October CHIRPS rainfall in the 2005 municipality below its 1981–2004 20th percentile (Jessoe et al., 2018). Remittances, a proxy, are cash received in the last 12 months from parents, siblings, or children outside the household. Expenditure, a proxy for income, is in 2002 pesos per person per month. Rural is a locality under 2,500 inhabitants (INEGI, 2024).

Table 3—Sample Sizes, Treated and Control Children

	Panel A: Binary DID, 2009 drought	
	Control	Treated
HAZ sample, children aged 15 or younger in 2005 and 2009		
Children, estimation sample	1,514	922
Children, Model 1 (basic DID)	1,512	920
Children, Model 3 (DID with controls)	1,077	678
Children, Model 5 (triple difference)	987	635
In 2002 remittance households	390	193
Households, estimation sample	944	578
2005 municipalities, estimation sample	87	46
WAZ sample, children aged 10 or younger in 2005 and 2009		
Children, estimation sample	696	388
Children, Model 1 (basic DID)	692	386
Children, Model 3 (DID with controls)	445	255
Children, Model 5 (triple difference)	400	234
In 2002 remittance households	168	90
Households, estimation sample	565	316
2005 municipalities, estimation sample	79	44
BMIZ sample, children aged 15 or younger in 2005 and 2009		
Children, estimation sample	1,487	896
Children, Model 1 (basic DID)	1,485	894
Children, Model 3 (DID with controls)	1,057	657
Children, Model 5 (triple difference)	968	619
In 2002 remittance households	381	189
Households, estimation sample	931	561
2005 municipalities, estimation sample	87	46

	Panel B: Continuous DID, any 2009 deficit	
	Control	Treated
HAZ sample, children aged 15 or younger in 2005 and 2009		
Children, continuous DID sample	829	1,607
In 2002 remittance households	236	438
In 2002 non-remittance households	437	738
Households	501	1,021
2005 municipalities	46	87
WAZ sample, children aged 10 or younger in 2005 and 2009		
Children, continuous DID sample	380	704
In 2002 remittance households	100	193
In 2002 non-remittance households	172	260
Households	304	577
2005 municipalities	42	81
BMIZ sample, children aged 15 or younger in 2005 and 2009		
Children, continuous DID sample	815	1,568
In 2002 remittance households	231	427
In 2002 non-remittance households	430	720
Households	496	996
2005 municipalities	46	87

	Panel C: Rainfall classes vs near normal	
	Control	Treated
Moderately dry, $-1.5 < z \leq -1$		
Children, HAZ sample	1,493	488
Children, WAZ sample	683	217
Children, BMIZ sample	1,464	472
2005 municipalities, HAZ sample	89	23
2005 municipalities, WAZ sample	80	23
2005 municipalities, BMIZ sample	89	23
Severely dry or worse, $z \leq -1.5$		
Children, HAZ sample	1,493	370
Children, WAZ sample	683	145
Children, BMIZ sample	1,464	364
2005 municipalities, HAZ sample	89	17
2005 municipalities, WAZ sample	80	16
2005 municipalities, BMIZ sample	89	17
Severely dry, $-2 < z \leq -1.5$		
Children, HAZ sample	1,493	302
Children, WAZ sample	683	121
Children, BMIZ sample	1,464	296
2005 municipalities, HAZ sample	89	15
2005 municipalities, WAZ sample	80	14
2005 municipalities, BMIZ sample	89	15
Extremely dry, $z \leq -2$		
Children, HAZ sample	1,493	68
Children, WAZ sample	683	24
Children, BMIZ sample	1,464	68
2005 municipalities, HAZ sample	89	2
2005 municipalities, WAZ sample	80	2
2005 municipalities, BMIZ sample	89	2

Notes: Counts are children, households, or 2005 municipalities in each regression sample, at the 2005 baseline. The estimation samples are children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 15 or younger for HAZ and BMIZ and 10 or younger for WAZ in both rounds. Children have a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. In Panel A, treated children's 2005 municipality had a drought in 2009, with May–October CHIRPS rainfall below its 1981–2004 20th percentile (Jessoe et al., 2018). Model 1 drops children alone in their interview-month cell, and the continuous DID, with no interview-month fixed effects, uses the full sample. Model 3 adds the 2005–2008 droughts and 2002 household and 2005 child and mother controls, and keeps children with all of them observed. Model 5 adds 2002 remittances times post and the drought times post times 2002 remittances, and keeps children with 2002 remittances observed, so Model 5 children not in remittance households are the Model 5 count minus the remittance row. In Panel B, treated children's 2005 municipality had 2009 rainfall below its 1981–2004 mean (z below 0), where z is 2009 rainfall in standard deviations from that mean. These are the continuous DID samples, and the two remittance rows are the groups of the remittance split. They do not add up to the total, as children with missing 2002 remittances are in neither row. Panel C compares each rainfall class with near normal (z between -1 and 1) in the continuous DID. The classes use standardized rainfall with the SPI cutoffs defined by McKee et al. and adopted by the World Meteorological Organization (McKee et al., 1993; WMO, 2012). Children in wet municipalities (z of 1 or above) are not in the rainfall class analysis. Remittances are cash received in the last 12 months from parents, siblings, or children living outside the household, from MxFLS-1 (2002).

Table 4—Sample Attrition Across Waves

	Number of children (N)		
	MxFLS-1 (2002)	MxFLS-2 (2005)	MxFLS-3 (2009)
HAZ sample, children under 16			
DID (2005 and 2009)		3,582	2,436
Event studies (all three waves)	2,723	1,856	1,300
WAZ sample, children under 11			
DID (2005 and 2009)		1,895	1,084
Event studies (all three waves)	962	609	275
BMIZ sample, children under 16			
DID (2005 and 2009)		3,511	2,383
Event studies (all three waves)	2,569	1,719	1,203

Notes: Counts of children in each wave. The DID only uses MxFLS-2 (2005) and MxFLS-3 (2009), and the event studies use all three waves. The continuous DID uses the same samples. Each row starts from the children who meet the regression sample rules in its first wave and could stay in the sample, not from all children measured. They are under the age cutoff in every wave, 15 or younger for HAZ and BMIZ and 10 or younger for WAZ, and have a municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Later columns count the children still in the sample, so the drop across columns is attrition. This is not the same as the children the regressions drop as singletons, with only one wave in the sample, as these also include children who aged out or are only in a later wave. The DID 2009 count is the estimation sample, and the event study 2009 count is the children in all three waves. For children not measured in a later wave, the age then comes from their birth date and their household's MxFLS-3 measurement date, as MxFLS-3 interviews ran over several years. Children not in MxFLS-3, or whose birth date does not agree with their reported age, get their last measured age plus the median age gap between the waves, 4 years from 2005 to 2009 and 8 years from 2002 to 2009. This is an estimate, but 92 and 86 percent of these gaps are within a year of the median among children measured in both waves, and a guess off by a year only matters for children near the age cutoff, who could be counted as attrition when they aged out or the other way around. So those who would have aged out anyway are not counted as attrition.

Regression Results

Table 5—2009 Drought and Child Anthropometrics, Rainfall Classes

	HAZ		WAZ		BMIZ	
	(1)	(2)	(3)	(4)	(5)	(6)
Dry, $z \leq -1$	-0.041		-0.197		-0.163**	
	(0.100)		(0.117)		(0.076)	
	[0.704]		[0.104]		[0.036]	
Moderately dry, $-1.5 < z \leq -1$		0.018		-0.230		-0.221*
		(0.139)		(0.164)		(0.106)
		[0.915]		[0.220]		[0.062]
Severely dry or worse, $z \leq -1.5$		-0.117		-0.148		-0.086
		(0.112)		(0.119)		(0.068)
		[0.308]		[0.238]		[0.218]
P-val: pre-trend 2002–2005, moderately dry		0.933		0.393		0.230
P-val: pre-trend 2002–2005, severely dry or worse		0.791		0.237		0.040
Children	2,351	2,351	1,045	1,045	2,300	2,300
Dry 2005 municipalities	40		39		40	
Near-normal 2005 municipalities	89	89	80	80	89	89

Notes: This table compares children in dry 2005 municipalities with children in near-normal ones. Regressions are at the child level, on the 2005 to 2009 change in the WHO z-score, for the binary DID estimation samples without children in wet municipalities (z of 1 or above). z is 2009 May–October CHIRPS rainfall, the maize growing season (Jessoe et al., 2018), in the child's 2005 municipality in standard deviations from its 1981–2004 mean, grouped into standardized rainfall classes with the SPI cutoffs: near normal is z between -1 and 1, moderately dry between -1.5 and -1, and severely dry or worse -1.5 or below (McKee et al., 1993). Cols. 1, 3 and 5 compare dry (z of -1 or below) with near normal, and Cols. 2, 4 and 6 split dry into moderately dry and severely dry or worse. Each class is only compared with near normal, which needs parallel trends, and not with the other class, which would need strong parallel trends (Callaway et al., 2024). Since the contdid package does not yet support discrete doses, the classes are estimated with the discrete-dose regression in equation 4.1 (Callaway et al., 2024). The regressions have no controls, as the contdid package does not yet support covariates (Callaway et al., 2024). The pre-trend p-values test the 2002 leads of the rainfall class event study, on the same children with child and wave fixed effects. Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 6—2009 Drought and BMI-for-Age, Binary DID

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.145**	-0.180**	-0.304***	-0.212*	-0.266**
	(0.071)	(0.087)	(0.095)	(0.112)	(0.110)
Drought × post × 2002 remittances					0.027
					(0.217)
2002 remittances × post					0.010
					(0.074)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	2,379	2,379	1,714	712	1,587

Notes: This table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID. Regressions are at the child-wave level. The sample is children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 15 or younger in both rounds. Cols. 3–5 also keep only children with all of their regressors observed. Children have a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Drought equals 1 if May–October CHIRPS rainfall in 2009, the maize growing season (Jessoe et al., 2018), in the child's 2005 municipality was below its 1981–2004 20th percentile, and post equals 1 in MxFLS-3. Remittances equal 1 if the household received cash in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. Cols. 1–4 report the average drought effect. Col. 1 is the basic DID, Col. 2 adds the 2005–2008 droughts, and Col. 3 adds 2002 household controls (electricity, floor, roof, ventilation, sleeping rooms per person, log expenditure per capita, rural) and 2005 child and mother controls (child sex and full vaccination, mother's height, schooling, age at birth and employment). Col. 4 adds the 2005 father and birth controls (father's height, schooling, age at birth and employment, birth order, oldest child, birth weight, delivered at home, weeks predominantly breastfed, iron in pregnancy, premature), which are missing for many children, often because the father migrated. All controls are interacted with post. Col. 5 is the triple difference: it keeps the controls of Col. 3 and adds 2002 remittances × post and drought × post × 2002 remittances, the difference in the drought effect for children in 2002 remittance households. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). All regressions include child and interview year-month fixed effects, children in singleton cells are dropped, and standard errors are clustered by 2005 municipality, where the drought is assigned. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 7—Continuous DID and Remittances as a Moderator

	HAZ (1)	WAZ (2)	BMIZ (3)
Panel A: Continuous DID, any 2009 rainfall deficit			
Average ATT	-0.077	0.028	-0.045
	(0.101)	(0.097)	(0.069)
	[0.462]	[0.777]	[0.518]
Average ACRT	-0.007	-0.033	0.029
	(0.137)	(0.173)	(0.130)
Children	2,436	1,084	2,383
Panel B: Average ATT by 2002 remittances			
No 2002 remittances	-0.080	0.050	-0.042
	(0.107)	(0.152)	(0.091)
2002 remittances	-0.279	-0.030	0.031
	(0.182)	(0.190)	(0.116)
Difference, remittances minus none	-0.199	-0.081	0.074
	(0.179)	(0.218)	(0.145)
Children	1,849	725	1,808
Panel C: Dry vs near normal by 2002 remittances			
Dry, $z \leq -1$	-0.054	-0.160	-0.251**
	(0.114)	(0.200)	(0.109)
	[0.657]	[0.543]	[0.027]
Dry × 2002 remittances	-0.135	-0.220	0.100
	(0.133)	(0.216)	(0.207)
	[0.317]	[0.329]	[0.653]
2002 remittances	-0.040	0.088	0.039
	(0.109)	(0.134)	(0.070)
	[0.722]	[0.524]	[0.577]
Children	1,773	693	1,734

Notes: This table reports the continuous DID (Callaway et al., 2024) and tests whether 2002 remittances moderated the effect of the 2009 drought. Regressions are at the child level, on the 2005 to 2009 change in the WHO z-score, for the binary DID estimation samples. The dose is the 2009 rainfall deficit, max(0, -z), where z is 2009 May–October CHIRPS rainfall, the maize growing season (Jessoe et al., 2018), in the child's 2005 municipality in standard deviations from its 1981–2004 mean. Panel A compares children with any 2009 deficit (z below 0) with children without one, Panel B gives the average ATT separately for children in 2002 remittance and non-remittance households and their difference, and Panel C interacts dry (z of -1 or below) vs near normal with 2002 remittances. Remittances are cash received in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. The average ATT needs parallel trends, while the average ACRT, the average slope of the dose response, and comparisons between doses need strong parallel trends (Callaway et al., 2024). Panel A uses the contdid package (Callaway et al., 2024). Panels B and C are regressions on the 2005 to 2009 change, and Panel C uses the discrete-dose regression in equation 4.1 (Callaway et al., 2024). The estimator does not allow controls or fixed effects. Standard errors in parentheses are clustered by 2005 municipality, where the drought is assigned. Because the package only bootstraps at the child level, we coded a multiplier bootstrap of its influence functions that clusters by 2005 municipality, and the other rows use municipality-clustered regression standard errors. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE 1—2009 RAINFALL BY MUNICIPALITY, RAINFALL CLASSES

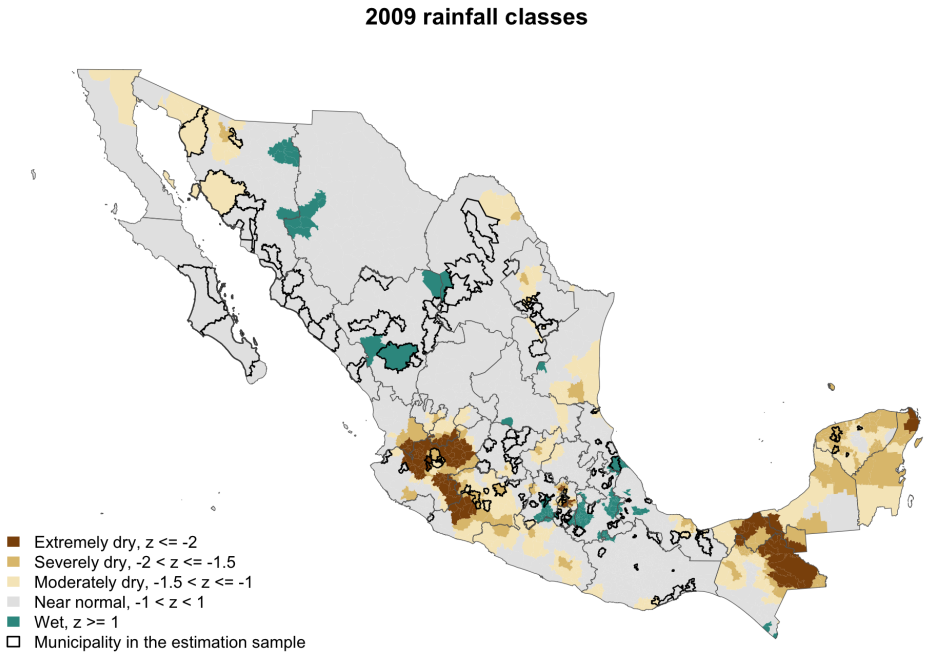


FIGURE 2—2009 DROUGHT BY MUNICIPALITY, BINARY DID TREATMENT

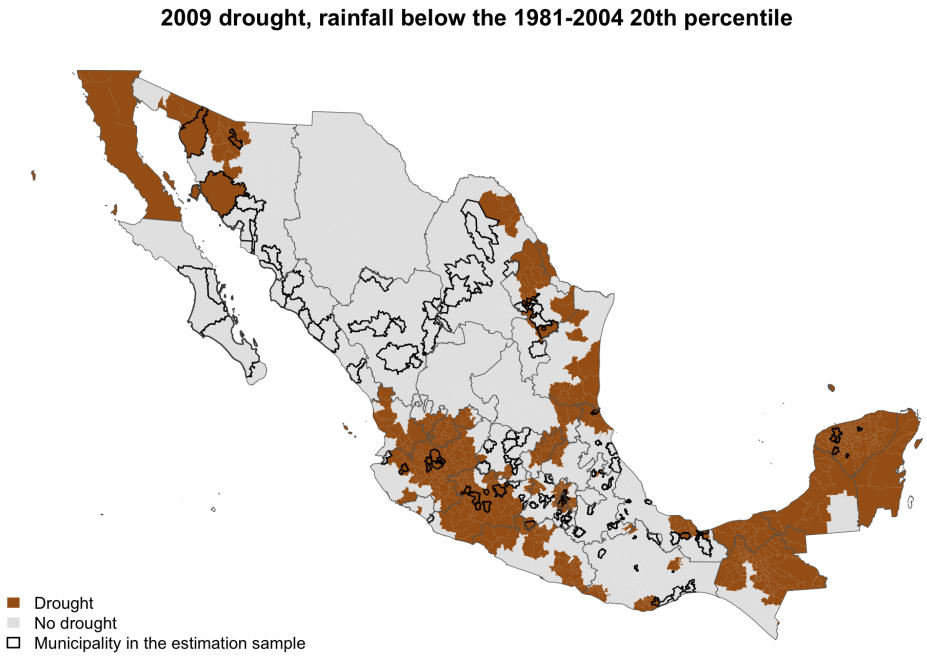
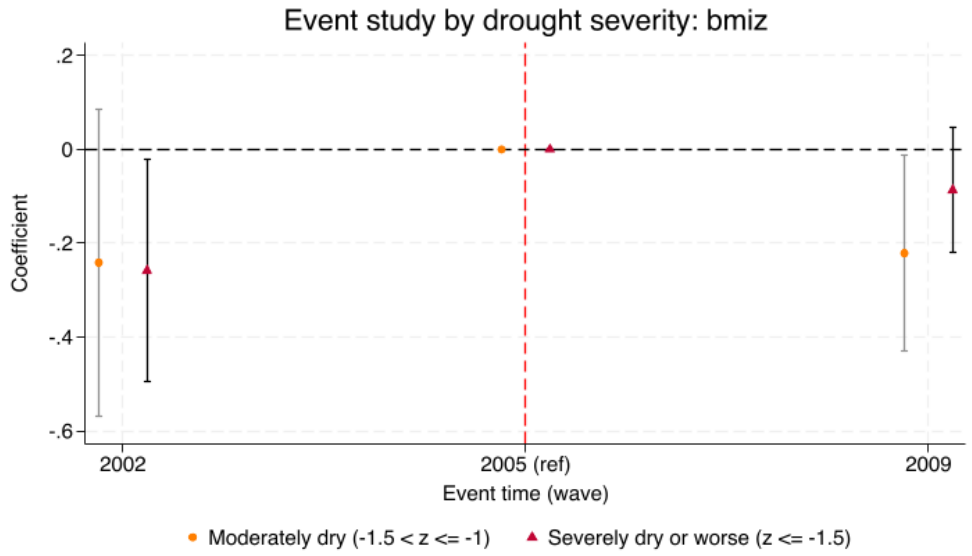
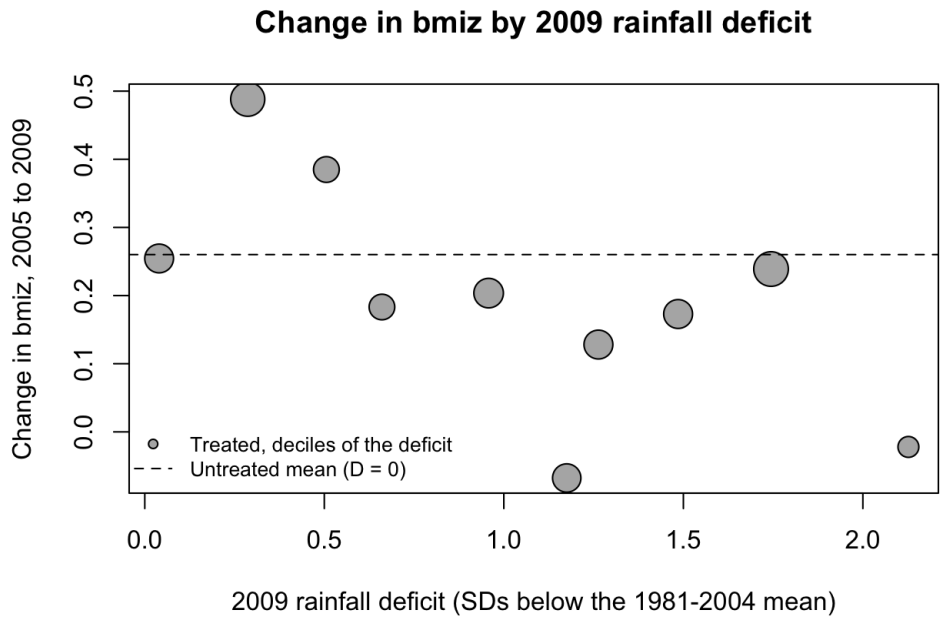


Figure 3—Event Study by Drought Severity, BMI-for-Age, Rainfall Classes



Notes: Moderately dry and severely dry or worse vs near normal, from one regression on the children of the rainfall class table, in each wave they are measured. No controls, same as the rainfall class table. Child and wave fixed effects, 2005 is the reference wave, and 95% confidence intervals use standard errors clustered by 2005 municipality.

Figure 4—Change in BMI-for-Age by 2009 Rainfall Deficit, Continuous DID



Appendix

Table 8—2009 Drought and Height-for-Age, Binary DID

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.072	-0.035	0.021	0.101	-0.005
	(0.096)	(0.103)	(0.113)	(0.168)	(0.118)
Drought × post × 2002 remittances					-0.153
					(0.148)
2002 remittances × post					-0.165*
					(0.090)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	2,432	2,432	1,755	734	1,622

Notes: This table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID. Regressions are at the child-wave level. The sample is children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 15 or younger in both rounds. Cols. 3–5 also keep only children with all of their regressors observed. Children have a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Drought equals 1 if May–October CHIRPS rainfall in 2009, the maize growing season (Jessoe et al., 2018), in the child's 2005 municipality was below its 1981–2004 20th percentile, and post equals 1 in MxFLS-3. Remittances equal 1 if the household received cash in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. Cols. 1–4 report the average drought effect. Col. 1 is the basic DID, Col. 2 adds the 2005–2008 droughts, and Col. 3 adds 2002 household controls (electricity, floor, roof, ventilation, sleeping rooms per person, log expenditure per capita, rural) and 2005 child and mother controls (child sex and full vaccination, mother's height, schooling, age at birth and employment). Col. 4 adds the 2005 father and birth controls (father's height, schooling, age at birth and employment, birth order, oldest child, birth weight, delivered at home, weeks predominantly breastfed, iron in pregnancy, premature), which are missing for many children, often because the father migrated. All controls are interacted with post. Col. 5 is the triple difference: it keeps the controls of Col. 3 and adds 2002 remittances × post and drought × post × 2002 remittances, the difference in the drought effect for children in 2002 remittance households. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). All regressions include child and interview year-month fixed effects, children in singleton cells are dropped, and standard errors are clustered by 2005 municipality, where the drought is assigned. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 9—2009 Drought and Weight-for-Age, Binary DID

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.200**	-0.196**	-0.299**	-0.181	-0.119
	(0.101)	(0.095)	(0.125)	(0.193)	(0.147)
Drought × post × 2002 remittances					-0.376*
					(0.205)
2002 remittances × post					0.096
					(0.119)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,078	1,078	700	224	634

Notes: This table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID. Regressions are at the child-wave level. The sample is children with the outcome measured in both MxFLS-2 (2005) and MxFLS-3 (2009), aged 10 or younger in both rounds. Cols. 3–5 also keep only children with all of their regressors observed. Children have a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Drought equals 1 if May–October CHIRPS rainfall in 2009, the maize growing season (Jessoe et al., 2018), in the child's 2005 municipality was below its 1981–2004 20th percentile, and post equals 1 in MxFLS-3. Remittances equal 1 if the household received cash in the last 12 months from parents, siblings, or children living outside the household in MxFLS-1 (2002), a proxy for remittances. Cols. 1–4 report the average drought effect. Col. 1 is the basic DID, Col. 2 adds the 2005–2008 droughts, and Col. 3 adds 2002 household controls (electricity, floor, roof, ventilation, sleeping rooms per person, log expenditure per capita, rural) and 2005 child and mother controls (child sex and full vaccination, mother's height, schooling, age at birth and employment). Col. 4 adds the 2005 father and birth controls (father's height, schooling, age at birth and employment, birth order, oldest child, birth weight, delivered at home, weeks predominantly breastfed, iron in pregnancy, premature), which are missing for many children, often because the father migrated. All controls are interacted with post. Col. 5 is the triple difference: it keeps the controls of Col. 3 and adds 2002 remittances × post and drought × post × 2002 remittances, the difference in the drought effect for children in 2002 remittance households. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). All regressions include child and interview year-month fixed effects, children in singleton cells are dropped, and standard errors are clustered by 2005 municipality, where the drought is assigned. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 10—2009 Drought and Child Anthropometrics, Binary DID Model 3 with All Covariates

	HAZ (1)	WAZ (2)	BMIZ (3)
2009 drought × post	0.021	-0.299**	-0.304***
	(0.113)	(0.125)	(0.095)
2005 drought × post	0.127	-0.193	-0.129
	(0.114)	(0.124)	(0.085)
2006 drought × post	-0.254	-0.343	-0.043
	(0.158)	(0.294)	(0.199)
2007 drought × post	-0.137	-0.181	-0.202*
	(0.118)	(0.152)	(0.120)
2008 drought × post	-0.070	0.037	0.350**
	(0.154)	(0.181)	(0.138)
2002 dwelling has electricity × post	-0.707**	-0.494**	0.016
	(0.321)	(0.211)	(0.239)
2002 good floor quality × post	-0.140	-0.031	0.137
	(0.114)	(0.156)	(0.102)
2002 good roof quality × post	0.083	-0.161	-0.155
	(0.093)	(0.160)	(0.099)
2002 enough ventilation × post	0.057	-0.010	0.140**
	(0.077)	(0.124)	(0.065)
2002 sleeping rooms per member × post	0.116	0.224	0.258*
	(0.187)	(0.255)	(0.146)
2002 ln expenditure per capita × post	0.023	0.147**	-0.044
	(0.053)	(0.064)	(0.042)
2002 rural locality × post	-0.137	-0.033	0.016
	(0.096)	(0.115)	(0.073)
Male × post	0.089	0.021	-0.000
	(0.069)	(0.103)	(0.061)
2005 mother's height (cm) × post	-0.009	-0.005	-0.001
	(0.006)	(0.008)	(0.005)
2005 mother's years of schooling × post	0.008	0.023	0.015
	(0.012)	(0.015)	(0.011)
Mother's age at birth × post	-0.002	0.005	-0.001
	(0.006)	(0.007)	(0.005)
2005 mother employed × post	-0.033	-0.171	-0.011
	(0.077)	(0.117)	(0.079)
2005 fully vaccinated × post	-0.020	0.282***	0.053
	(0.086)	(0.105)	(0.064)
Child FE	Y	Y	Y
Interview year-month FE	Y	Y	Y
Children	1,755	700	1,714
2005 municipalities (clusters)	116	101	116

Notes: This table reports every coefficient of Model 3, Col. 3 of the main binary DID tables, for each outcome. Samples, variables and controls are as in those tables, and all controls are interacted with post. All regressions include child and interview year-month fixed effects, children in singleton cells are dropped, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 11—2009 Drought and Height-for-Age, Binary DID, All Three Waves

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	0.023	0.048	0.144	0.099	0.181*
	(0.087)	(0.090)	(0.090)	(0.136)	(0.103)
Drought × post × 2002 remittances					-0.166
					(0.141)
2002 remittances × post					-0.084
					(0.092)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,293	1,293	1,084	559	1,003

Notes: This robustness check keeps only children measured in all three waves (2002, 2005 and 2009), so the event study pre-trend test and the 2005–2009 DID come from the same children, and changes in sample composition, such as attrition or children first measured in 2005, cannot drive the estimates. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for these children, aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 12—2009 Drought and Weight-for-Age, Binary DID, All Three Waves

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	0.025	-0.084	0.064	0.361	0.293
	(0.136)	(0.176)	(0.183)	(0.266)	(0.233)
Drought × post × 2002 remittances					-0.444
					(0.360)
2002 remittances × post					0.149
					(0.257)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	266	266	203	121	179

Notes: This robustness check keeps only children measured in all three waves (2002, 2005 and 2009), so the event study pre-trend test and the 2005–2009 DID come from the same children, and changes in sample composition, such as attrition or children first measured in 2005, cannot drive the estimates. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for these children, aged 10 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 13—2009 Drought and BMI-for-Age, Binary DID, All Three Waves

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.147	-0.223**	-0.250**	-0.171	-0.190
	(0.089)	(0.107)	(0.118)	(0.121)	(0.147)
Drought × post × 2002 remittances					-0.032
					(0.247)
2002 remittances × post					0.012
					(0.098)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,196	1,196	1,006	513	933

Notes: This robustness check keeps only children measured in all three waves (2002, 2005 and 2009), so the event study pre-trend test and the 2005–2009 DID come from the same children, and changes in sample composition, such as attrition or children first measured in 2005, cannot drive the estimates. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for these children, aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 14—2009 Drought and Height-for-Age, Binary DID, Boys

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.123	-0.108	-0.108	0.228	-0.075
	(0.123)	(0.130)	(0.146)	(0.237)	(0.142)
Drought × post × 2002 remittances					-0.301
					(0.257)
2002 remittances × post					-0.254**
					(0.123)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,201	1,201	869	353	806

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for boys, using sex reported in the 2005 household roster, aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 15—2009 Drought and Weight-for-Age, Binary DID, Boys

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.219	-0.175	-0.264	-0.330	-0.064
	(0.153)	(0.149)	(0.182)	(0.358)	(0.196)
Drought × post × 2002 remittances					-0.225
					(0.283)
2002 remittances × post					0.084
					(0.179)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	551	551	367	103	331

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for boys, using sex reported in the 2005 household roster, aged 10 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 16—2009 Drought and BMI-for-Age, Binary DID, Boys

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.170*	-0.168	-0.269**	-0.262	-0.336*
	(0.099)	(0.119)	(0.131)	(0.180)	(0.175)
Drought × post × 2002 remittances					0.416
					(0.322)
2002 remittances × post					-0.057
					(0.111)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,170	1,170	847	346	787

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for boys, using sex reported in the 2005 household roster, aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 17—2009 Drought and Height-for-Age, Binary DID, Girls

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.046	0.020	0.111	-0.055	0.035
	(0.101)	(0.116)	(0.131)	(0.202)	(0.139)
Drought × post × 2002 remittances					-0.060
					(0.176)
2002 remittances × post					-0.143
					(0.115)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,220	1,220	878	377	811

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for girls, using sex reported in the 2005 household roster, aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 18—2009 Drought and Weight-for-Age, Binary DID, Girls

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.259**	-0.282**	-0.379**	-0.184	-0.265
	(0.111)	(0.119)	(0.167)	(0.288)	(0.197)
Drought × post × 2002 remittances					-0.401
					(0.302)
2002 remittances × post					0.097
					(0.189)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	520	520	324	115	295

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for girls, using sex reported in the 2005 household roster, aged 10 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 19—2009 Drought and BMI-for-Age, Binary DID, Girls

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.112	-0.160	-0.313***	-0.200	-0.184
	(0.077)	(0.097)	(0.116)	(0.167)	(0.114)
Drought × post × 2002 remittances					-0.342*
					(0.200)
2002 remittances × post					0.065
					(0.098)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,199	1,199	861	364	795

Notes: Estimating the effect separately for boys and girls tests for gender disparities in the impact of the drought, as households facing an income shock may protect the nutrition and health of sons and daughters differently. The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for girls, using sex reported in the 2005 household roster, aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without child sex in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 20—2009 Drought and Height-for-Age, Binary DID, Rural Children

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.045	0.189	0.190	0.472	0.273*
	(0.162)	(0.173)	(0.163)	(0.289)	(0.159)
Drought × post × 2002 remittances					-0.278
					(0.288)
2002 remittances × post					-0.006
					(0.130)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,016	1,016	732	272	714

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for rural children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 21—2009 Drought and Weight-for-Age, Binary DID, Rural Children

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.178	-0.074	-0.276	0.532	-0.131
	(0.180)	(0.172)	(0.264)	(0.334)	(0.281)
Drought × post × 2002 remittances					-0.362
					(0.332)
2002 remittances × post					0.216
					(0.225)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	414	414	265	85	260

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for rural children (2005 household locality), aged 10 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 22—2009 Drought and BMI-for-Age, Binary DID, Rural Children

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.096	-0.138	-0.208	-0.244	-0.197
	(0.112)	(0.127)	(0.180)	(0.254)	(0.204)
Drought × post × 2002 remittances					-0.035
					(0.255)
2002 remittances × post					0.086
					(0.095)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	998	998	719	264	701

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for rural children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 23—2009 Drought and Height-for-Age, Binary DID, Urban Children

	Height-for-age z-score (HAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.102	-0.136	-0.088	-0.021	-0.243*
	(0.106)	(0.110)	(0.125)	(0.200)	(0.122)
Drought × post × 2002 remittances					0.032
					(0.167)
2002 remittances × post					-0.331***
					(0.121)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,412	1,412	1,020	455	904

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO height-for-age z-score (HAZ) with the binary DID for urban children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 24—2009 Drought and Weight-for-Age, Binary DID, Urban Children

	Weight-for-age z-score (WAZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.282**	-0.370***	-0.446***	-0.213	-0.201
	(0.114)	(0.108)	(0.135)	(0.254)	(0.175)
Drought × post × 2002 remittances					-0.486*
					(0.261)
2002 remittances × post					0.035
					(0.145)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	654	654	427	137	365

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO weight-for-age z-score (WAZ) with the binary DID for urban children (2005 household locality), aged 10 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 25—2009 Drought and BMI-for-Age, Binary DID, Urban Children

	BMI-for-age z-score (BMIZ)				
	(1)	(2)	(3)	(4)	(5)
Drought × post	-0.189**	-0.259***	-0.382***	-0.192	-0.316**
	(0.078)	(0.080)	(0.096)	(0.141)	(0.128)
Drought × post × 2002 remittances					0.043
					(0.327)
2002 remittances × post					0.049
					(0.107)
2005–2008 droughts × post	N	Y	Y	Y	Y
2002 household controls × post	N	N	Y	Y	Y
2005 child and mother controls × post	N	N	Y	Y	Y
2005 father and birth controls × post	N	N	N	Y	N
Child FE	Y	Y	Y	Y	Y
Interview year-month FE	Y	Y	Y	Y	Y
Children	1,377	1,377	992	441	882

Notes: Estimating the effect separately for rural and urban children tests whether the drought hits harder where household incomes depend more directly on rain-fed agriculture, while urban households are mainly exposed through food prices. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). The table reports the effect of the 2009 drought on the WHO BMI-for-age z-score (BMIZ) with the binary DID for urban children (2005 household locality), aged 15 or younger in both rounds, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Columns, variables and controls are as in the main binary DID tables, without rural in the controls. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 26—Event Study, 2009 Drought and Height-for-Age

	Height-for-age z-score (HAZ)	
	(1) Event study	(2) Triple difference
Drought × 2002 wave (lead)	0.115	0.192
	(0.116)	(0.128)
Drought × 2009 wave	0.015	-0.008
	(0.110)	(0.114)
Drought × 2002 remittances × 2002 wave (lead)		-0.297
		(0.180)
Drought × 2002 remittances × 2009 wave		-0.158
		(0.147)
2005–2008 droughts × wave	Y	Y
2002 household controls × wave	Y	Y
2005 child and mother controls × wave	Y	Y
Child FE	Y	Y
Interview year-month FE	Y	Y
P-value, lead = 0	0.322	0.101
2009 drought effect, remittance households		-0.167
SE		0.143
P-value		0.245
Observations (child-waves)	4,598	4,249
Children	1,755	1,622
2005 municipalities (clusters)	116	114

Notes: This table tests for pre-trends before the 2009 drought. Regressions are at the child-wave level on MxFLS-1 (2002), MxFLS-2 (2005) and MxFLS-3 (2009), with 2005 as the reference wave. The sample is children aged 15 or younger in each wave they are in, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Col. 1 keeps the children in Model 3 of the binary DID and Col. 2 those in Model 5, so each matches its DID sample. Col. 1 regresses the outcome on the drought times the 2002 wave, known as the lead, and times the 2009 wave, with the 2005–2008 droughts, 2002 household and 2005 child and mother controls, each times the 2002 and 2009 waves. Following the literature, baseline controls are interacted with time to allow different trends by pre-treatment characteristics (Bailey and Goodman-Bacon, 2015; Bailey et al., 2021). Col. 2 adds the drought terms times 2002 remittances, and the 2002 remittance terms on their own (not shown). The P-value of the lead tests the drought lead in Col. 1 and the triple difference lead in Col. 2. The effect for remittance households is the sum of the two 2009 drought terms. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 27—Event Study, 2009 Drought and Weight-for-Age

	Weight-for-age z-score (WAZ)	
	(1) Event study	(2) Triple difference
Drought × 2002 wave (lead)	-0.481**	-0.336
	(0.216)	(0.244)
Drought × 2009 wave	-0.298**	-0.105
	(0.126)	(0.145)
Drought × 2002 remittances × 2002 wave (lead)		-0.078
		(0.347)
Drought × 2002 remittances × 2009 wave		-0.373*
		(0.208)
2005–2008 droughts × wave	Y	Y
2002 household controls × wave	Y	Y
2005 child and mother controls × wave	Y	Y
Child FE	Y	Y
Interview year-month FE	Y	Y
P-value, lead = 0	0.028	0.822
2009 drought effect, remittance households		-0.477
SE		0.200
P-value		0.019
Observations (child-waves)	1,609	1,454
Children	700	634
2005 municipalities (clusters)	101	99

Notes: This table tests for pre-trends before the 2009 drought. Regressions are at the child-wave level on MxFLS-1 (2002), MxFLS-2 (2005) and MxFLS-3 (2009), with 2005 as the reference wave. The sample is children aged 10 or younger in each wave they are in, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Col. 1 keeps the children in Model 3 of the binary DID and Col. 2 those in Model 5, so each matches its DID sample. Col. 1 regresses the outcome on the drought times the 2002 wave, known as the lead, and times the 2009 wave, with the 2005–2008 droughts, 2002 household and 2005 child and mother controls, each times the 2002 and 2009 waves. Following the literature, baseline controls are interacted with time to allow different trends by pre-treatment characteristics (Bailey and Goodman-Bacon, 2015; Bailey et al., 2021). Col. 2 adds the drought terms times 2002 remittances, and the 2002 remittance terms on their own (not shown). The P-value of the lead tests the drought lead in Col. 1 and the triple difference lead in Col. 2. The effect for remittance households is the sum of the two 2009 drought terms. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 28—Event Study, 2009 Drought and BMI-for-Age

	BMI-for-age z-score (BMIZ)	
	(1) Event study	(2) Triple difference
Drought × 2002 wave (lead)	-0.387***	-0.423***
	(0.104)	(0.120)
Drought × 2009 wave	-0.293***	-0.262**
	(0.097)	(0.114)
Drought × 2002 remittances × 2002 wave (lead)		0.186
		(0.202)
Drought × 2002 remittances × 2009 wave		0.040
		(0.217)
2005–2008 droughts × wave	Y	Y
2002 household controls × wave	Y	Y
2005 child and mother controls × wave	Y	Y
Child FE	Y	Y
Interview year-month FE	Y	Y
P-value, lead = 0	0.000	0.358
2009 drought effect, remittance households		-0.222
SE		0.167
P-value		0.187
Observations (child-waves)	4,438	4,110
Children	1,714	1,587
2005 municipalities (clusters)	116	114

Notes: This table tests for pre-trends before the 2009 drought. Regressions are at the child-wave level on MxFLS-1 (2002), MxFLS-2 (2005) and MxFLS-3 (2009), with 2005 as the reference wave. The sample is children aged 15 or younger in each wave they are in, with a 2005 municipality. They were not measured in MxFLS-3 before November 2009. We use November 2009 as the cutoff because the 2009 rainfall deficit, the drought, had not fully occurred before then. Dropping them from both groups equalizes treated and control measurement timing. Col. 1 keeps the children in Model 3 of the binary DID and Col. 2 those in Model 5, so each matches its DID sample. Col. 1 regresses the outcome on the drought times the 2002 wave, known as the lead, and times the 2009 wave, with the 2005–2008 droughts, 2002 household and 2005 child and mother controls, each times the 2002 and 2009 waves. Following the literature, baseline controls are interacted with time to allow different trends by pre-treatment characteristics (Bailey and Goodman-Bacon, 2015; Bailey et al., 2021). Col. 2 adds the drought terms times 2002 remittances, and the 2002 remittance terms on their own (not shown). The P-value of the lead tests the drought lead in Col. 1 and the triple difference lead in Col. 2. The effect for remittance households is the sum of the two 2009 drought terms. All regressions include child and interview year-month fixed effects, and standard errors are clustered by 2005 municipality. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 29—Continuous DID by Subgroup, Height-for-Age

	Change in the height-for-age z-score (HAZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	-0.050	-0.108	0.101	-0.180
	(0.114)	(0.116)	(0.181)	(0.101)
	[0.672]	[0.376]	[0.614]	[0.104]
Dry vs near normal, $z \leq -1$	-0.073	-0.009	0.033	-0.095
	(0.114)	(0.115)	(0.180)	(0.106)
	[0.533]	[0.937]	[0.876]	[0.406]
Severely dry or worse vs near normal, $z \leq -1.5$	-0.158	-0.077	0.181	-0.241*
	(0.158)	(0.106)	(0.191)	(0.115)
	[0.344]	[0.488]	[0.519]	[0.065]
Dry × 2002 remittances	-0.364	0.089	-0.197	-0.098
	(0.226)	(0.168)	(0.211)	(0.175)
	[0.121]	[0.610]	[0.375]	[0.586]
Children	1,208	1,228	1,020	1,416
Municipalities with no 2009 deficit	42	41	27	24

Notes: This table repeats the continuous DID and rainfall class results for the WHO height-for-age z-score (HAZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought, using sex reported in the 2005 household roster, and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). Rows, samples and definitions are as in the main continuous DID and rainfall class tables. Children are aged 15 or younger in both rounds. The any deficit row uses the contdid package, and the rainfall class rows use the discrete-dose regression in equation 4.1 (Callaway et al., 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 30—Continuous DID by Subgroup, Weight-for-Age

	Change in the weight-for-age z-score (WAZ)			
	Boys (1)	Girls (2)	Rural (3)	Urban (4)
Any 2009 deficit	0.139	-0.089	0.066	0.047
	(0.143)	(0.112)	(0.202)	(0.094)
	[0.348]	[0.436]	[0.756]	[0.622]
Dry vs near normal, $z \leq -1$	-0.221	-0.179	-0.143	-0.218**
	(0.180)	(0.111)	(0.227)	(0.103)
	[0.262]	[0.122]	[0.639]	[0.044]
Severely dry or worse vs near normal, $z \leq -1.5$	-0.152	-0.153	0.013	-0.237
	(0.198)	(0.170)	(0.124)	(0.154)
	[0.471]	[0.428]	[0.924]	[0.168]
Dry × 2002 remittances	-0.008	-0.456	-0.144	-0.327
	(0.301)	(0.274)	(0.348)	(0.250)
	[0.977]	[0.102]	[0.705]	[0.209]
Children	556	528	424	660
Municipalities with no 2009 deficit	36	32	23	23

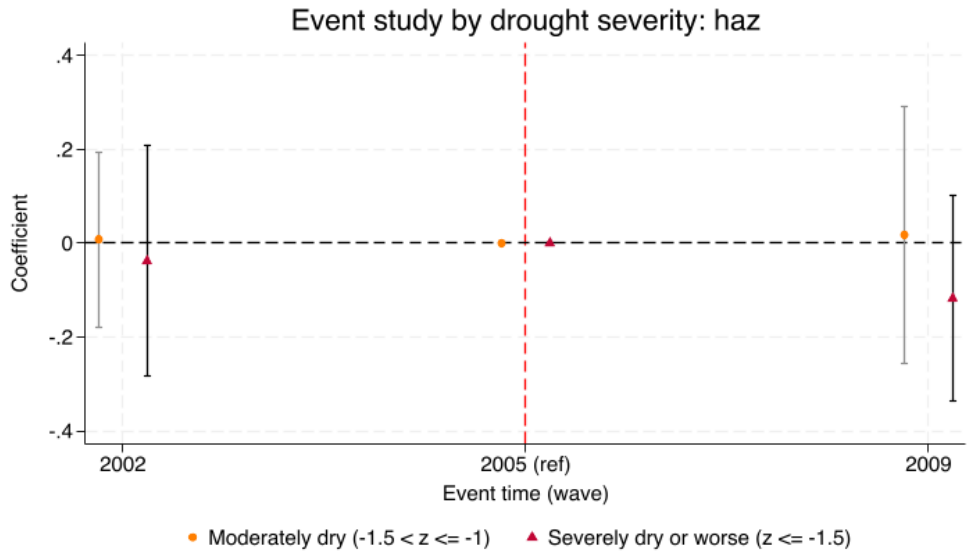
Notes: This table repeats the continuous DID and rainfall class results for the WHO weight-for-age z-score (WAZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought, using sex reported in the 2005 household roster, and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). Rows, samples and definitions are as in the main continuous DID and rainfall class tables. Children are aged 10 or younger in both rounds. The any deficit row uses the contdid package, and the rainfall class rows use the discrete-dose regression in equation 4.1 (Callaway et al., 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 31—Continuous DID by Subgroup, BMI-for-Age

	Change in the BMI-for-age z-score (BMIZ)			
	Boys	Girls	Rural	Urban
	(1)	(2)	(3)	(4)
Any 2009 deficit	-0.008	-0.082	-0.035	-0.054
	(0.093)	(0.076)	(0.090)	(0.095)
	[0.933]	[0.294]	[0.699]	[0.593]
Dry vs near normal, $z \leq -1$	-0.202*	-0.124	-0.147	-0.174
	(0.101)	(0.080)	(0.094)	(0.111)
	[0.054]	[0.138]	[0.155]	[0.141]
Severely dry or worse vs near normal, $z \leq -1.5$	-0.064	-0.106	-0.129	-0.077
	(0.105)	(0.102)	(0.090)	(0.085)
	[0.548]	[0.350]	[0.309]	[0.379]
Dry × 2002 remittances	0.506	-0.301	-0.030	0.216
	(0.324)	(0.180)	(0.263)	(0.304)
	[0.142]	[0.119]	[0.915]	[0.534]
Children	1,177	1,206	1,002	1,381
Municipalities with no 2009 deficit	42	41	27	24

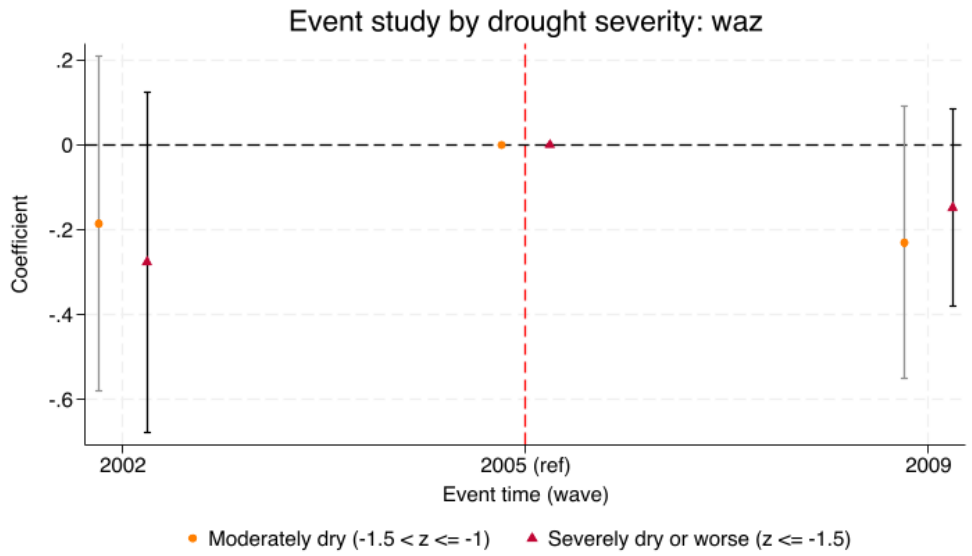
Notes: This table repeats the continuous DID and rainfall class results for the WHO BMI-for-age z-score (BMIZ) by subgroup. Cols. 1–2 test for gender disparities in the impact of the drought, using sex reported in the 2005 household roster, and Cols. 3–4 test whether it hits rural children harder, as rural household incomes depend more directly on rain-fed agriculture. Rural is a locality with fewer than 2,500 inhabitants (INEGI, 2024). Rows, samples and definitions are as in the main continuous DID and rainfall class tables. Children are aged 15 or younger in both rounds. The any deficit row uses the contdid package, and the rainfall class rows use the discrete-dose regression in equation 4.1 (Callaway et al., 2024). Standard errors clustered by 2005 municipality are in parentheses. Wild cluster bootstrap p-values are in brackets, which tend to be conservative with few treated municipalities (MacKinnon et al., 2023). Stars use the clustered p-values, and where a wild bootstrap p-value is shown both must pass. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Figure 5—Event Study by Drought Severity, Height-for-Age, Rainfall Classes



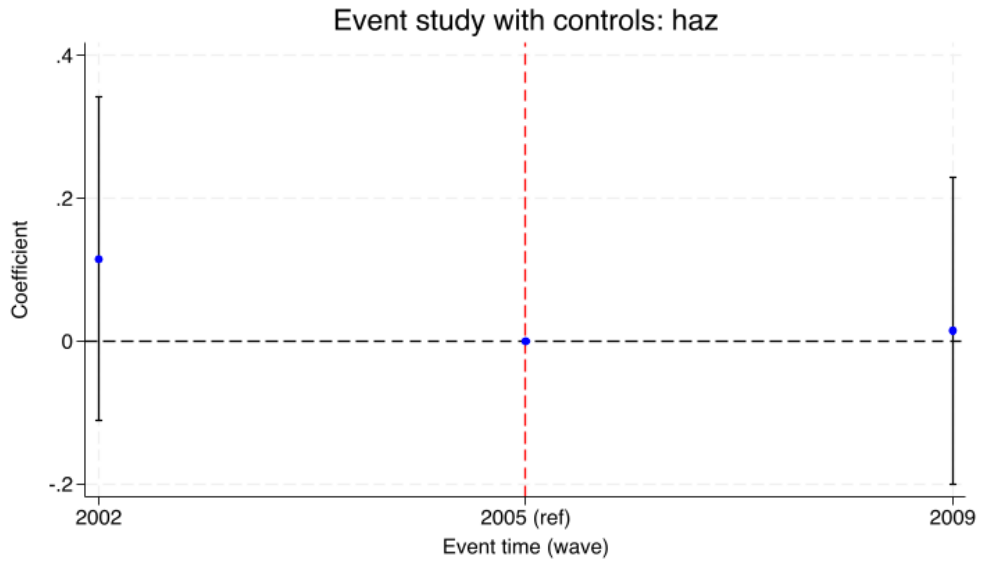
Notes: Moderately dry and severely dry or worse vs near normal, from one regression on the children of the rainfall class table, in each wave they are measured. No controls, same as the rainfall class table. Child and wave fixed effects, 2005 is the reference wave, and 95% confidence intervals use standard errors clustered by 2005 municipality.

Figure 6—Event Study by Drought Severity, Weight-for-Age, Rainfall Classes



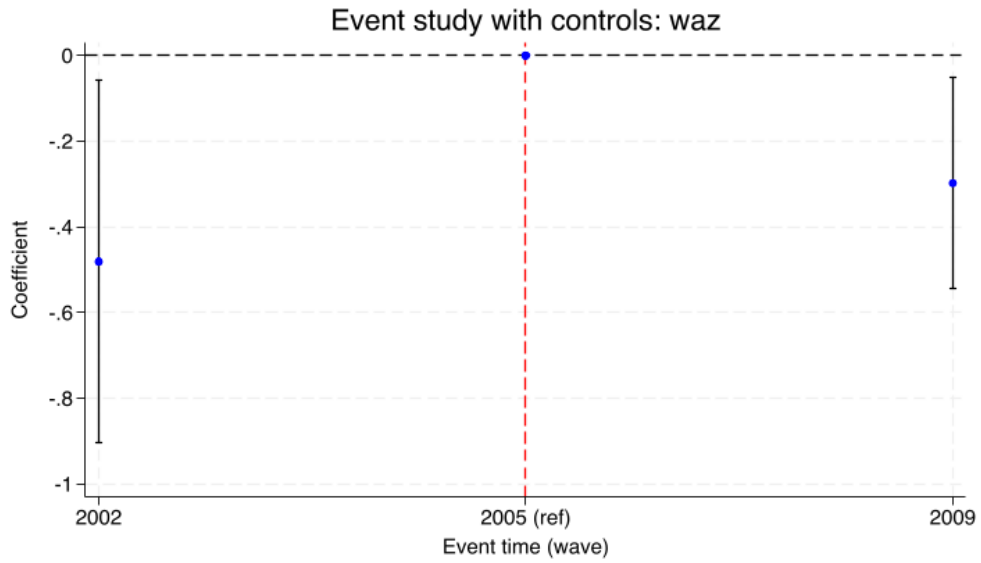
Notes: Moderately dry and severely dry or worse vs near normal, from one regression on the children of the rainfall class table, in each wave they are measured. No controls, same as the rainfall class table. Child and wave fixed effects, 2005 is the reference wave, and 95% confidence intervals use standard errors clustered by 2005 municipality.

Figure 7—Event Study, Height-for-Age, Binary DID, With Controls



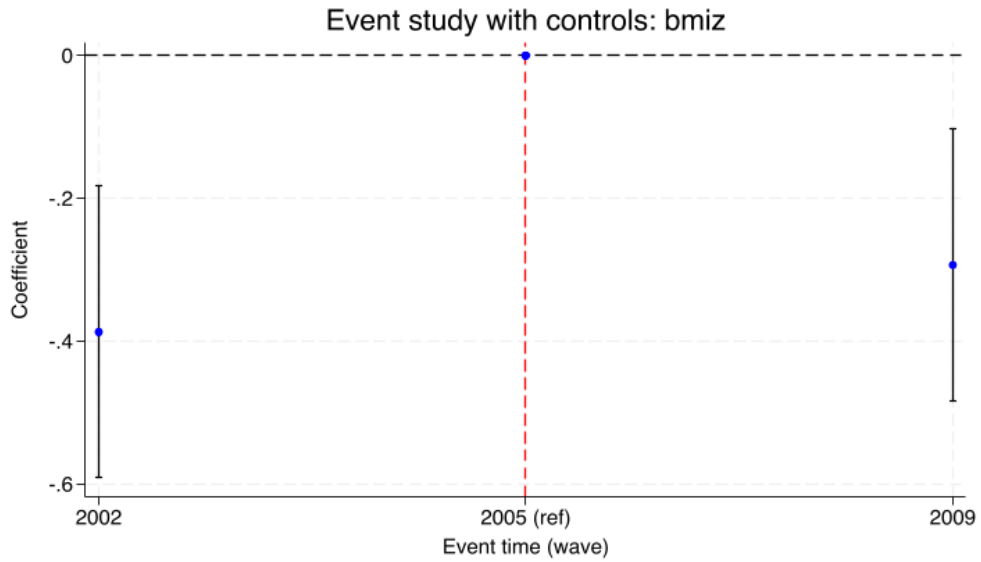
Notes: Sample is the children in Model 3 of the binary DID, in each wave they are measured. The Model 3 controls are each interacted with the 2002 and 2009 waves. Child and interview year-month fixed effects, 2005 is the reference wave, and 95% confidence intervals use standard errors clustered by 2005 municipality.

Figure 8—Event Study, Weight-for-Age, Binary DID, With Controls



Notes: Sample is the children in Model 3 of the binary DID, in each wave they are measured. The Model 3 controls are each interacted with the 2002 and 2009 waves. Child and interview year-month fixed effects, 2005 is the reference wave, and 95% confidence intervals use standard errors clustered by 2005 municipality.

FIGURE 9—EVENT STUDY, BMI-FOR-AGE, BINARY DID, WITH CONTROLS



Notes: Sample is the children in Model 3 of the binary DID, in each wave they are measured. The Model 3 controls are each interacted with the 2002 and 2009 waves. Child and interview year-month fixed effects, 2005 is the reference wave, and 95% confidence intervals use standard errors clustered by 2005 municipality.

FIGURE 10—CHANGE IN HEIGHT-FOR-AGE BY 2009 RAINFALL DEFICIT, CONTINUOUS DID

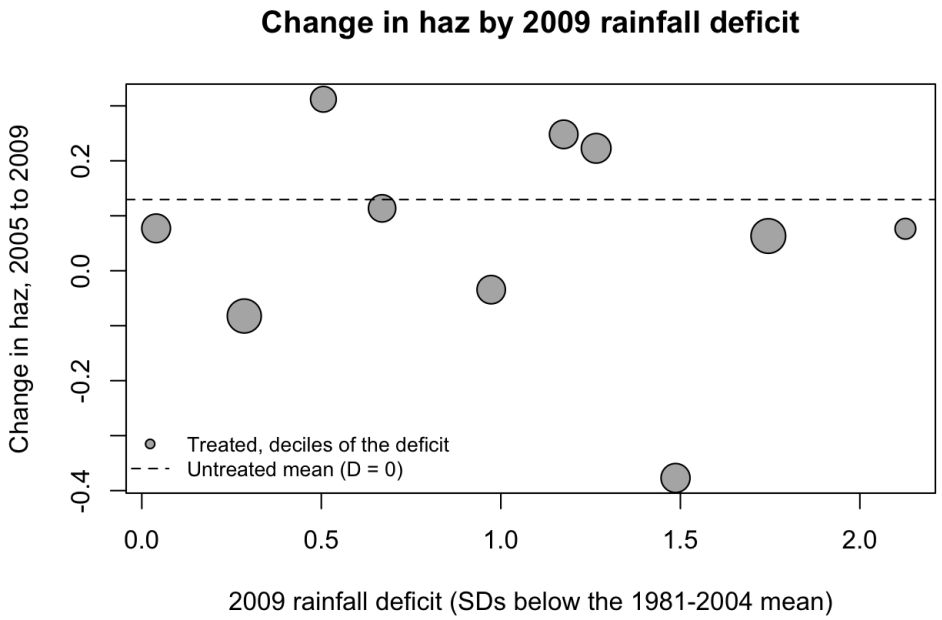


Figure 11—Change in Weight-for-Age by 2009 Rainfall Deficit, Continuous DID

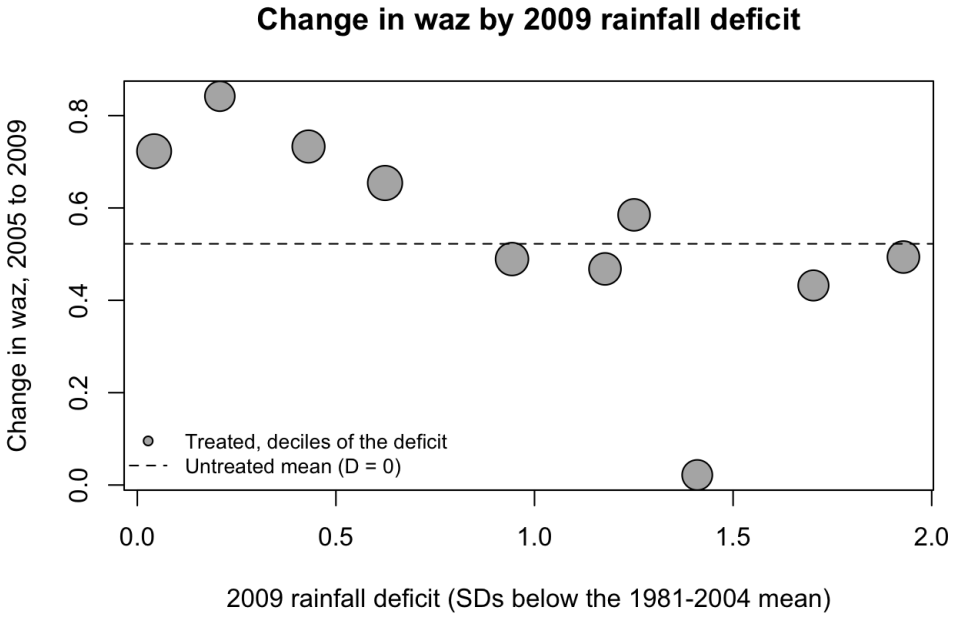


Figure 12—ATT by 2009 Rainfall Deficit, Height-for-Age, Continuous DID

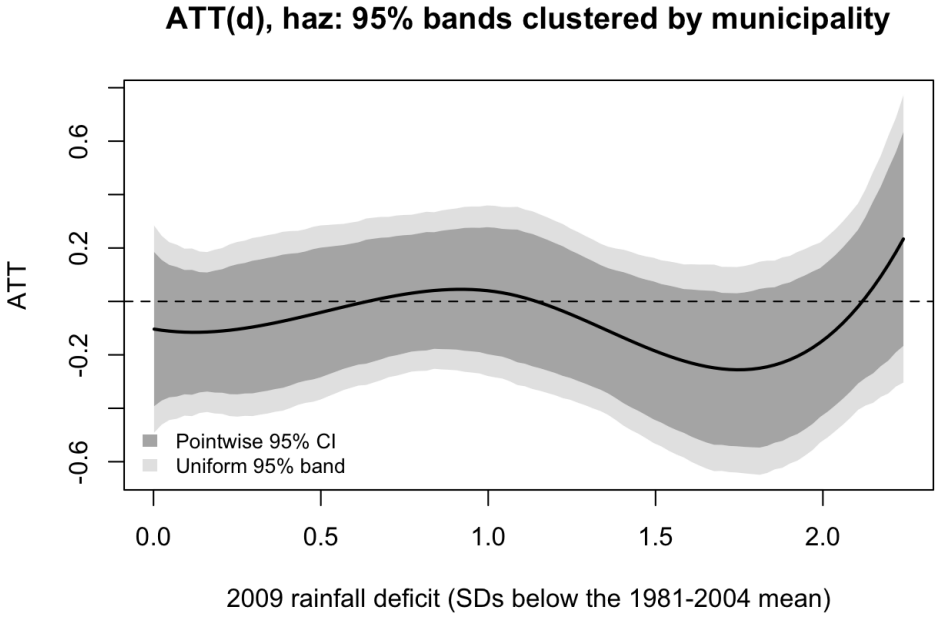


FIGURE 13—ATT BY 2009 RAINFALL DEFICIT, WEIGHT-FOR-AGE, CONTINUOUS DID

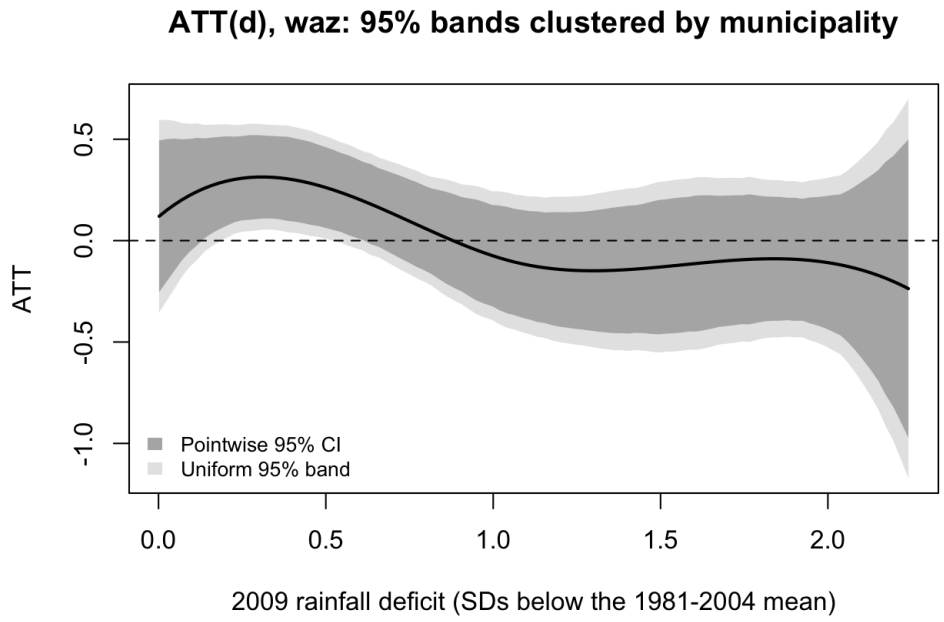


FIGURE 14—ATT BY 2009 RAINFALL DEFICIT, BMI-FOR-AGE, CONTINUOUS DID

